

Arbitrarily large ranks over $\mathbb{Q}$

Problèmes ouverts — A. Dujella · 2026

Problem 2.1 (Dujella, 2026). Are there elliptic curves over $\mathbb{Q}$ of arbitrarily large rank?

Remark. The current record over $\mathbb{Q}$ is rank at least 29 (Elkies-Klagsbrun, 2024). Heuristics of Park, Poonen, Voight and Wood (2019) suggest that ranks over $\mathbb{Q}$ might be bounded.